

Library System Dude: Augmented Library Search through LLM and Knowledge Graphs

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1. Introduction

Library System Dude (LSD) is the culminating project for the 2024/25 edition of the Semantic Digital Library course taught by Professor Giovanni Colavizza at Alma Mater Studiorum in Bologna. Developed by team members Romolo David D'Alessandro, Valentino Castagna, and Niccolò Eros Molinati, LSD is a prototype for an agent that enables smart querying of a KG database via an intelligent agent – our Library System Dude.

2. Motivation and Objectives

Digital libraries have long provided high-quality information and various retrieval services. However, recent advancements in knowledge graphs and LLMs open new possibilities for enhancing these capabilities. In particular, modeling library holdings as a KG enables an LLM-based QA system to produce accurate answers that are directly grounded in structured data.

Several implementations have already explored this synergy. For example, Xu et al.¹ merged vectorized retrieval, KGs, and LLMs to create a balanced system in which each component compensates for the others' limitations. Ma et al.² proposed a hybrid retrieval-augmented generation (RAG) framework that iteratively extracts information from both unstructured and structured sources. Building on these foundations, our objective was to design and implement a prototype QA system in which RAG leverages a KG to guide SPARQL queries based on entities detected in user questions.

In our envisioned use case, a user seeks detailed information about topics covered by the digital library's resources. This interaction operates as follows:

- An entity linking (EL) tool identifies entities within the user's query.
- SPARQL queries leverage the KG to retrieve relevant documents associated with those entities.
- The retrieved documents are provided, along with the user's question, to an LLM that generates an answer.
- The generated answer includes a concise summary and references to the documents used.

To instantiate this pipeline, we employed an EL tool for entity detection, a KG that describes resources and associated entities, a graph database to store and query the KG, and an LLM to formulate answers. We implemented the entire pipeline using Python scripts and GraphDB software, aiming for a modular, lightweight framework that can be extended or improved over time.

¹ Ma, S., Xu, C., Jiang, X., Li, M., Qu, H., Yang, C., Mao, J., & Guo, J. (2024). Think-on-Graph 2.0: Deep and faithful large language model reasoning with knowledge-guided retrieval augmented generation. arXiv.

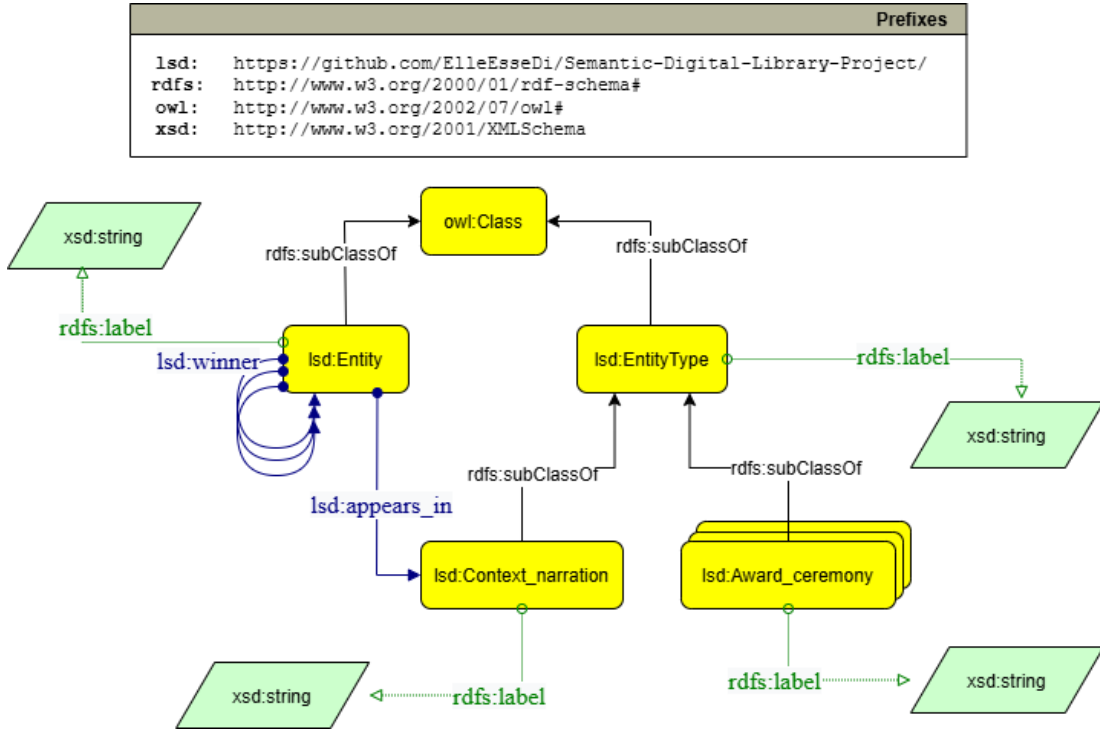
² Xu, L., Lu, L., & Zhang, Y. (2023). Nanjing Yunjin intelligent question-answering system based on knowledge graphs and retrieval augmented generation technology. ResearchGate.

3. Ontology Development

For our KG, we selected the EventNarrative³ dataset, a knowledge graph-to-text collection that comprises approximately 230,000 graphs paired with their corresponding natural language descriptions. EventNarrative provides entities linked to textual narratives, enabling high-quality annotations. The dataset is publicly available and offers a diverse set of event-centric records. Practically, our input was the dataset’s JSON file containing entities and their contexts.

3.1. The LSD Ontology

We opted for a lightweight ontology model to map our data, driven by two considerations: we wanted a highly navigable model that enabled fast retrieval without complex case-by-case scenarios, and we sought to provide a broad textual base upon which the LLM could build responses. Our ontology comprises two main classes: *Entity*, which contains all entities identified from the dataset and *EntityType*, the parent class for all types as well as a custom class, *ContextNarration*, assigned to contextual fragments. Since generating fully expressive ontology was not our primary goal, we focused on a model that is both comprehensible to new users and easily queried with a single SPARQL statement. The advantage of such simplicity is that the ontology can be generated automatically from the dataset, accommodating high type variance and large scale without manual modeling. We hosted the resulting KG in GraphDB (v. 10.8)⁴ due to its flexibility and free license.



4. Entity Linker

To detect entities within user questions, we evaluated various EL solutions based on scalability, multilingual capability, and performance. Although more powerful, scalable entity linkers such as Falcon 2.0 or BELA were considered beyond our prototypical scope. Instead, we selected the Radboud Entity Linker (REL), a user-friendly Python package with a web API that leverages English Wikipedia as its knowledge base. While REL’s limitation to English is noted – rendering it less effective for multilingual queries – it sufficed for our

³ Anthony Colas, Ali Sadeghian, Yue Wang, Daisy Wang. EventNarrative dataset. University of Florida.

⁴ What is GraphDB? GraphDB 10.8 documentation. Ontotext. <https://graphdb.ontotext.com/>

demonstrative purposes. In our prototype, we accessed REL⁵ via its web API, submitting the user’s question text and receiving a list of detected entities. These identified entities then serve as inputs to our graph handler script, which formulates and executes SPARQL queries to retrieve relevant KG resources.

5. LLM Agent

The LLM agent represents the final component in our KG-augmented QA system. We selected DeepSeek R1, a state-of-the-art reasoning model designed for augmented generation tasks. DeepSeek R1⁶ offers several advantages: its advanced reasoning capabilities facilitate the integration of content from multiple contexts, yielding more coherent outputs; employs a Chain-of-Thought (CoT) reasoning process that enables self-validation and error correction within generated outputs; was released alongside distilled versions that can be executed locally with limited computational resources, allowing us to operate the system fully offline if desired. Using DeepSeek R1, we ensured a flexible yet powerful LLM module, configurable for both remote API calls to leverage the largest available models and local inference via quantized versions.

5.2. Settings

In line with best practices for RAG systems, we configured the LLM settings to maintain determinism and prevent deviation from provided contexts:

- **max_tokens = 10000**: Sets the upper limit on tokens generated per response, controlling output length.
- **temperature = 0.3**: Reduces randomness by focusing probability on high-likelihood tokens.
- **top_k = 40**: Restricts token selection to the top 40 most probable candidates, further ensuring output consistency.
- **top_p = 0.8**: Applies nucleus sampling by selecting tokens whose cumulative probability reaches 80%, then sampling within that set.

These parameters collectively reduce the risk of hallucination and ensure that generated answers remain anchored to retrieved contexts.

5.3. LLM Calls

For remote inference, we utilized OpenRouter’s free-tier API, which grants limited access to various open-source models, including DeepSeek R1. Our implementation followed the OpenRouter chat completion API structure⁷, distinguishing between a system prompt and two user prompt. For local inference, we integrated Ollama. We selected the quantized 8-billion-parameter version of DeepSeek R1. Ollama’s client interface abstracts the chat call structure, thereby facilitating seamless transitions between remote and local modes.

5.4. Prompt Engineering

Our system prompt was crafted using fundamental prompt engineering strategies, including element-based and markup-based instructions. The first segment specifies the persona the LLM should assume – providing contextual background and a clear mission (for example, “You are Library System Dude, an expert in answering user queries about library resources”). The second segment lists strict instructions in two categories:

⁵ REL: Radboud Entity Linker. (n.d.). GitHub. <https://github.com/informagi/REL>.

⁶ DeepSeek-AI, Guo, D., Yang, D., Zhang, H., Song, J., Zhang, R., Xu, R., Zhu, Q., Ma, S., Wang, P., Bi, X., Zhang, X., Yu, X., Wu, Y., Wu, Z. F., Gou, Z., Shao, Z., Li, Z., Gao, Z., ... Zhang, Z. (2025). DeepSeek-R1: Incentivizing reasoning capability in LLMs via reinforcement learning. arXiv

⁷ OpenRouter. <https://openrouter.ai/>

- **Content Generation:** Directs the model to focus on relevant information from provided contexts and to produce a structured, cohesive answer.
- **Fringe Case Handling:** Instructs the model on scenarios of missing, insufficient, or incompatible information. If no contexts are available, the LLM should output a “NO INFO” string to indicate an inability to answer.

An internal method within our LLMAgentInterface class formats the input data – encompassing both the query and retrieved contexts – using XML-style tags. This method also handles the injection of “NO INFO” when necessary.

6. Testing

For testing, we resorted to use competency questions to test whether or not the LLM agent correctly follows instructions and to compare performances between remote and local models. For demonstrational purposes, we include some examples.

TEST 0: Can the LSD gracefully handle false information included in the query?

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#####TEST 0#####
Tell me about the stock-market crash happened in 2001

***Remote Answer***
I do not have any information available about a stock market crash specifically occurring in 2001. My knowledge on this topic is limited to the data provided, and there is no context available to address this query. If you have further details or a different question, feel free to ask!

***Local Answer***
The stock market crash that occurred in 2001 was primarily influenced by the events of September 11, 2001. This incident led to a significant decline in global stock markets, with the Dow Jones Industrial Average experiencing a substantial drop, losing over 500 points in one day. Investors panicked due to fear and uncertainty, causing widespread sell-offs not only in U.S. markets but also affecting other international exchanges. This crash underscores the profound impact that terrorist attacks can have on financial markets and investor confidence.
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TEST 1: Does LSD perform well with comparative queries?

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#####TEST 1#####
What are the Pacific Games and the Olympic games? Perform a comparative analysis of the two.

***Remote Answer***
The Pacific Games and the Olympic Games are both multi-sport events but differ significantly in scope, participation, and historical context, as evidenced by the available information:

### **Pacific Games**
- **Regional Focus**:: Exclusively involves Pacific Island nations and territories (e.g., Wallis and Futuna, American Samoa, Niue, New Caledonia, Samoa).
- **Sports**:: Features regional and traditional sports such as **outrigger canoeing**, **netball**, and **rugby union**, alongside mainstream sports like football, sailing, and volleyball. For example, outrigger canoeing events were held in 2007 and 2011, while netball competitions date back to 1966.
- **Frequency and History**:: Held quadrennially, with editions noted from 1966 to 2015. The 2011 Games in Nouméa, New Caledonia, included 185 competitors from Wallis and Futuna and 159 from American Samoa.
- **Scale**:: Smaller in scale, with fewer participating nations and athletes (e.g., Niue sent 63 competitors in 2011).

### **Olympic Games**
- **Global Participation**:: Includes nations worldwide (e.g., USA, China, Germany, Norway) across Summer and Winter editions. Events like the 2020 Tokyo Olympics and 2018 Pyeongchang Winter Olympics are highlighted.
- **Sports Diversity**:: Covers a vast range of sports, from **athletics**, **swimming**, and **cycling** to **alpine skiing**, **figure skating**, and **taekwondo**. Demonstration sports like curling (1992), and sledge (1994) are also noted.
- **Historical Depth**:: Dates back to the early 20th century (e.g., 1904 Summer Olympics, 1924 Winter Olympics) and includes iconic moments such as the 1992 "Dream Team" in basketball and Usain Bolt's 2016 100m victory.
- **Scale and Prestige**:: Far larger, with events like the 2016 Rio Olympics featuring 128 taekwondo athletes from 60 nations and complex qualification systems across sports.

### **Key Differences**
1. **Geographic Scope**:: Pacific Games are regionally confined; Olympics are global.
2. **Sports Profile**:: Pacific Games include culturally specific sports (e.g., outrigger canoeing), while the Olympics emphasize universal disciplines with historical milestones (e.g., Thorpe's decathlon in 1912, Pellegri's swimming records in 2008).
3. **Historical Context**:: The Olympics have a longer, more documented history, including political dimensions (e.g., Germany's exclusion in 1920-1948) and legacy events like the torch relay.

### **Limitations**
The data lacks details on governance, funding, or exact founding years for the Pacific Games, and direct comparative metrics (e.g., audience size, budgets). However, the synthesis confirms the Pacific Games' role as a regional platform versus the Olympics' global prestige and diversity.

***Local Answer***
The text describes the performances of various figure skating pairs at the 2006 Winter Olympics. It highlights the achievements and standings in the pairs event, noting:
1. **Gold Medal**:: Tatiana Totmianina/Maxim Marinin (Russia), who were two-time World Champions and reigning Olympic bronze medalists.
2. **Silver Medal**:: Zhang Dan/Zhang Hao (China), known for their performance to Led Zeppelin's "Kashmir."
3. **Bronze Medal**:: Maria Petrova/Alexei Tikhonov (Russia).

Other notable performances include:
- Rena Inoue/John Baldwin (USA) who made history with the first throw triple Axel in Olympic competition, finishing sixth.
- Pang Qing/Tong Jian (China), who narrowly missed the podium by 0.24 points.
- Shen Xue/Zhao Hongbo (China), despite an injury to Zhao, remained competitive.

The text also mentions that Aljona Savchenko/Robin Szolkowy (Germany) improved from seventh after the short program to fifth in free skating.

### Key Takeaways:
- **Competition Structure**:: The event included short and free skating programs, with the pairs' short program starting first.
- **Significant Achievements**:: Inoue/Baldwin's throw triple Axel and Totmianina/Marinin's victory were notable highlights.
- **National Rivalry**:: China had a strong presence with multiple top-tier performances.

This information provides insights into the competitive dynamics, technical achievements, and national representation at the 2006 Winter Olympics figure skating pairs event.
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TEST 2: Can LSD handle queries not including any explicit entities?

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####TEST 2####
What was the most brutal terrorist attack in the latest century?

***Remote Answer***
Based on the provided information, I do not have access to any data about terrorist attacks or related historical events to answer your query. The available information explicitly state s ***NO INFO*** on this topic.

***Local Answer***
The information provided does not contain any details about the most brutal terrorist attacks in the latest century. Therefore, I cannot provide a definitive answer based on the available context.

Answer: I don't have enough information to determine the most brutal terrorist attack of the latest century.

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TEST 3: Can LSD handle queries including named entities which are not present in the knowledge base?

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####TEST 3####
Who is Osama Bin Laden? What was his nationality?

***Remote Answer***
I do not have any information about Osama Bin Laden or his nationality in the provided knowledge sources.

***Local Answer***
Osama Bin Laden was a founder of the terrorist organization al-Qaeda and was associated with various extremist activities. He was born in Saudi Arabia, but there is some uncertainty regarding his nationality. While it's commonly stated that he was a Saudi national, there are also reports suggesting Yemeni descent, leading to conflicting information about his nationality. Therefore, it's accurate to note that he was born in Saudi Arabia, though his nationality remains uncertain due to these discrepancies.

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TEST 4: Does LSD perform well on specific topics?

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####TEST 4####
Tell me about Japanese art

***Remote Answer***
Japanese art encompasses diverse styles and historical influences. A key development emerged during the Meiji period (1868-1912) with the introduction of *Yōga* (Western-style painting), formalized through institutions like the Kobu Bijutsu Gakkō (1876). While *Yōga* initially declined due to a resurgence of traditional *Nihonga* (Japanese-style painting), it regained prominence after 1990. Notably, *Nihonga* itself integrated Western techniques like perspective and shading, blurring the distinction between the two styles.

In the 19th century, *Japonism*—the Western fascination with Japanese art—profoundly influenced European artists, particularly through *ukiyo-e* prints, following Japan's reopening to trade in 1858.

Contemporary Japanese art is exemplified by figures like Takashi Murakami, whose 2012 exhibition *Murakami-Ego* in Qatar showcased his fusion of traditional motifs with pop culture, reflecting modern artistic innovation.

***Limitations***: The provided information does not address other facets of Japanese art (e.g., sculpture, ceramics, pre-Meiji traditions). Further details on specific movements, artists, or cultural contexts would require additional sources.

***Local Answer***
The list of historical events related to Japan provides a comprehensive overview of the country's rich and complex history, spanning from ancient times to more recent post-war developments. Here is a structured and organized presentation of these events, arranged chronologically:

1. **Battle of Danghangpo (Unknown - 13th Century)**: A naval engagement during the Japanese invasions of Korea, resulting in a Korean victory.
2. **Minamoto no Yoshitsune (12th Century)**: A prominent samurai who became an Ashikaga retainer after being exiled but later returned to aid the shogunate against their enemies.
3. **Ashikaga Shogunate (1185-1571)**: The rise of the Ashikaga family as shoguns, leading to significant political and military changes in Japan.
4. **Yakusho (active 17th century)**: A Japanese painter known for ink wash landscapes, reflecting the cultural vibrancy during the Edo period.
5. **Japan-Korea Annexation Treaty (1904)**: Japan annexed Korea under this treaty, leading to increased tensions and eventual global conflict.
6. **Surrender in WWII (August 14, 1945)**: Emperor Hirohito's announcement of surrender, ending World War II and leading to occupation by Allied forces.
7. **Constitution of Japan (May 3, 1947)**: Post-WWII document influenced by the American model, redefining the role of the Emperor and emphasizing human rights.
8. **Return of Okinawa (1956)**: Okinawa was returned to Japanese control after being administered by U.S. military forces since WWII.
9. **Joint Declaration with Soviet Union (1956)**: Transfer of Habomai Islands and Shikotan to Japan, still a point of contention today.
10. **House of Councillors Elections (1953)**: A step in Japan's post-war political restructuring, reflecting the shift towards democratic governance.
11. **Battle of Nanchang (1938)**: A significant WWII battle where Chinese forces resisted Japanese invasions, highlighting the war's impact on China.
12. **General Elections**: - 1890, 1892, 1953, 1993, and 2010 reflect Japan's evolving political landscape, adapting to internal and external pressures over time.
13. **Japonism (Late 19th-Early 20th Century)**: A cultural movement where Western art was influenced by Japanese aesthetics, post-Meiji Restoration.
14. **Minori (Modern Era)**: A visual novel company contributing to Japan's modern media culture and international influence.

This timeline illustrates the historical trajectory of Japan, from ancient governance through periods of military conflict, political transformation, and cultural exchange, highlighting its role in shaping modern global dynamics.

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Test	Remote	Local
TEST 0: Can the LSD gracefully handle false information included in the query?	Y	N
TEST 1: Does LSD perform well with comparative queries?	Y	N
TEST 2: Can LSD handle queries not including any explicit entities?	Y	Y
TEST 3: Can LSD handle queries including named entities which are not present in the knowledge base?	Y	N
TEST 4: Does LSD perform well on specific topics?	U	N
Y(es) N(o) U(ndecisive)		

Main takeaways

- Without surprise, the local model showed significant limitations in contrast to the remote one with regards to text generation and cohesion. An important point, is the incapacity to orchestrate different info into one complete answer;

- Overall, the remote model seems successful in producing exhaustive info and organize it in clear-cut sections, following instructions specified in the query;
- In tests 4 and 1, LSD mentioned limitations of the retrieved information, even though it is not supposed to know information outside the one present in the KB. This could suggest the necessity of more strict directives in terms of information leaking.

7. Conclusions

Our work serves as a proof of concept for a lightweight, modular framework that integrates LLM and KG interactions to enhance database search capabilities. Diverging from conventional RAG models – which often rely on vector embeddings of texts – LSD depends on a comprehensive knowledge graph, drawing inspiration from the TOG-2 study. While our pipeline demonstrates feasibility, it remains inefficient and imprecise in knowledge extraction. We identified main paths for improvement:

- **Multilingual Entity Linking:** Presently, our model performs optimally with English prompts. Incorporating a multilingual EL tool (for example, BELA or Falcon 2.0) would broaden accessibility and improve performance across languages.
- **Expanding the Search Mechanism:** Implementing a recursive, iterative search – where initial query results are evaluated and, if relevance falls below a threshold, the query expands to include proximate contexts – could enhance coverage and answer accuracy. This approach aligns with methods explored in the TOG-2 study.
- **Introducing vector representation as fallback option:** Following Xu et al.⁸, we could implement a vector database containing our documents as a secondary option in case graph queries fail. This would allow for a more resilient and adaptable model, while still maintaining the advantages of the KG layer.

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